so if $A = 8 = (1000)_2$
$B = 7 = (0111)_2$

multipy A by each bit of B
$B = b_3 b_2 b_1 b_0 = 0111$ respectively

pp0 = A and b_0 = 1000
pp1 = A and b_1 = 1000
pp2 = A and b_2 = 1000
pp3 = A and b_3 = 0000

product[0] = 0
r0 = {0, pp[3], pp[2], pp[1]}
~~= {0, 1, 0, 0~~
= 0100

pp1 + r0 = 1000 carry = 0
0100
1100

product[1] = 0
r1 = {0, 0110}

pp2 + r1 = 1000 carry = 0
0110
1110

product[2] = 0
r2 = 0111

pp3+r2 = 0000 carry = 0

+ 0111

0111

product [7:3] = 00111

product = 00111000

$(00111000)_2 = 8 + 16 + 32$

$= 56$

→ 8 x 7 = 56 Hence proved